

Task 39 — Explore Graph, Axis, and Chart

Objective: understand how data is visually represented and how the x-axis, y-axis, and chart types communicate trends, comparisons, proportions, and distributions.

4. What is a Graph?

Definition: A graph is a visual representation of data or a mathematical relationship. It makes patterns and relationships easier to see.

A graph can use points, lines, bars, or other visual forms depending on the type of information being represented.

5. X-axis and Y-axis

Definition: The x-axis is the horizontal axis. The y-axis is the vertical axis.

- X-axis → commonly represents input, category, time, or an independent variable.
- Y-axis → commonly represents output, measured value, or a dependent variable.
- A coordinate is commonly written as (x, y) .
- The exact meaning depends on the data being represented.

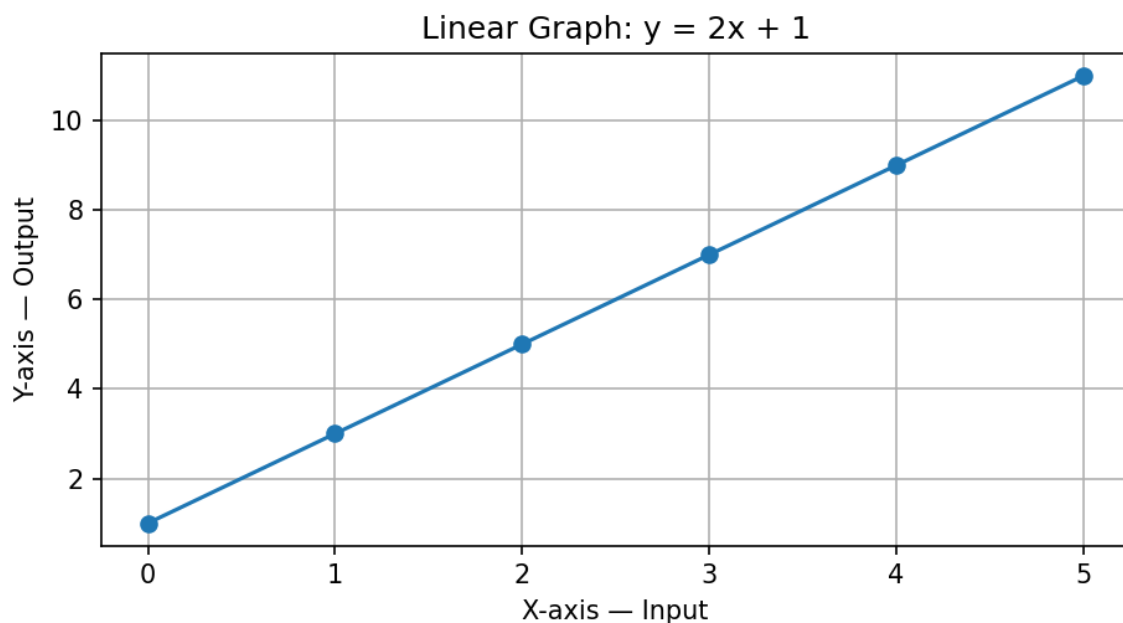
Example

```
x = 3
y = 2 * x + 1
print(y)
```

Output: 7

6. Linear Graph — $y = 2x + 1$

Definition: A linear graph shows a linear relationship. In $y = 2x + 1$, x is the input and y is the output; the line shows how the output changes as the input changes.



Observation: X represents input, Y represents output, and the line represents their relationship.

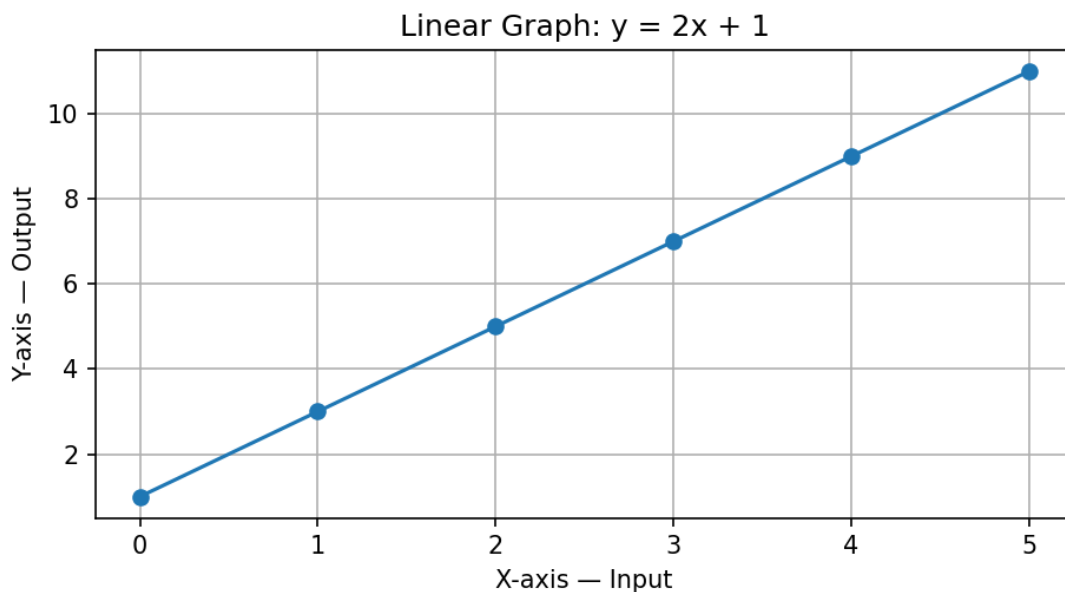
7. Line Chart

Definition: A line chart connects data points to show a trend or change, especially across an ordered variable such as time.

Example

```
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr']
sales = [20, 35, 28, 45]
plt.plot(months, sales, marker='o')
plt.xlabel('Month'); plt.ylabel('Sales'); plt.title('Monthly Sales')
plt.show()
```

Output: A line chart showing sales changing from January to April.



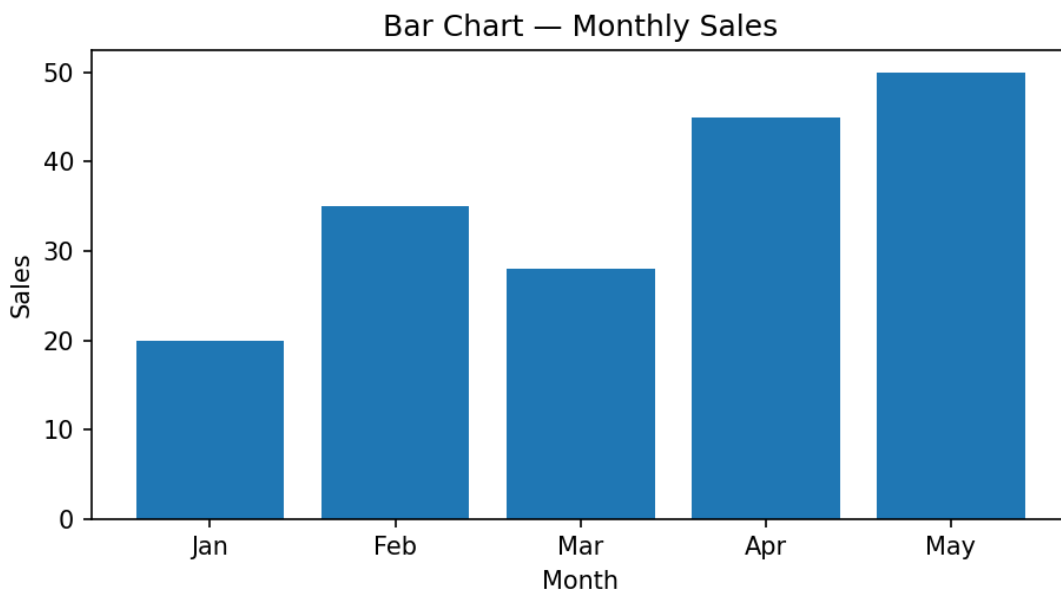
8. Bar Chart

Definition: A bar chart uses separate bars to compare quantities across categories.

Example

```
import matplotlib.pyplot as plt
subjects = ['Python', 'Pandas', 'NumPy']
marks = [80, 70, 90]
plt.bar(subjects, marks)
plt.xlabel('Subject'); plt.ylabel('Marks'); plt.title('Marks by Subject')
plt.show()
```

Output: Three bars comparing marks; NumPy has the highest value.



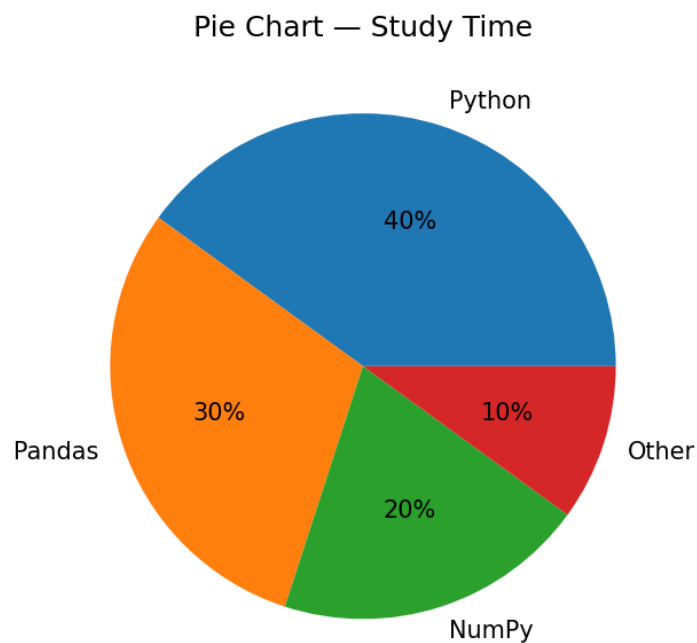
9. Pie Chart

Definition: A pie chart represents parts of a whole using sectors of a circle. The complete circle represents 100%.

Example

```
import matplotlib.pyplot as plt
labels = ['Python', 'Pandas', 'NumPy', 'Other']
values = [40, 30, 20, 10]
plt.pie(values, labels=labels, autopct='%1.0f%%')
plt.show()
```

Output: A circular chart showing each category's percentage of the total.



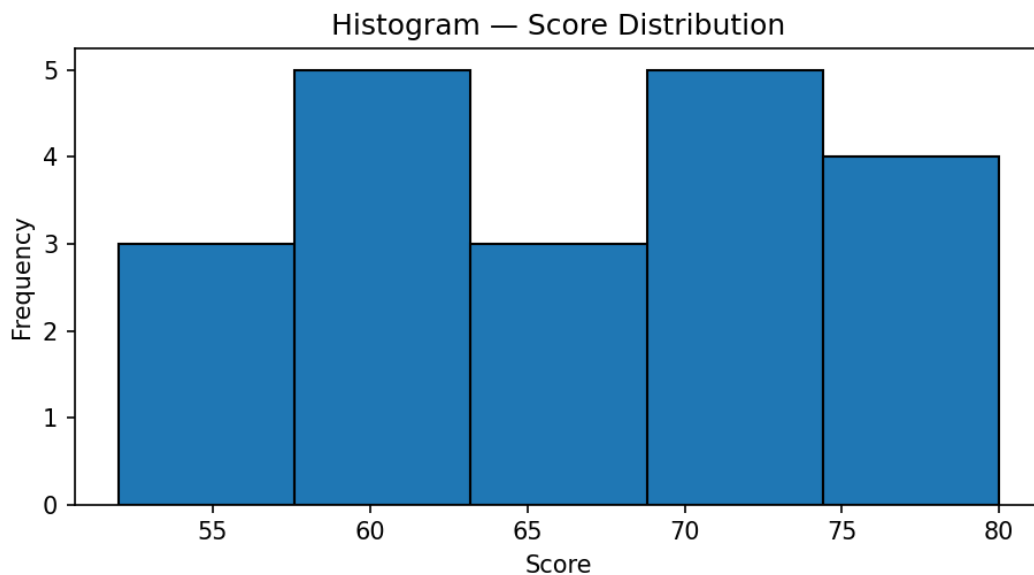
10. Histogram

Definition: A histogram groups numerical values into intervals called bins and displays how frequently values occur in those intervals.

Example

```
import matplotlib.pyplot as plt
scores = [52,55,57,60,61,62,63,65,66,68,70,71,72,74,75,76,78,80]
plt.hist(scores, bins=5)
plt.xlabel('Score'); plt.ylabel('Frequency')
plt.show()
```

Output: A histogram showing how scores are distributed across numerical intervals.



11. Comparing Chart Types

Chart	Purpose	Typical X-axis	Y-axis / Meaning
Line chart	Show trends/change	Time or ordered variable	Measured value
Bar chart	Compare categories	Categories	Quantity
Pie chart	Show parts of a whole	Usually none	Proportion shown by sectors
Histogram	Show numerical distribution	Numeric bins	Frequency/count

12. How Variables Are Placed on Axes

Definition: When using an x-y graph, choose axes based on what the variables represent. Often the input, category, or ordered variable is placed on the x-axis and the corresponding measured output is placed on the y-axis.

- Time vs temperature → X = time, Y = temperature → line chart.
- Department vs employees → X = department, Y = employee count → bar chart.
- Score distribution → X = score intervals, Y = frequency → histogram.
- Budget allocation → categories and percentages → pie chart.